

Aim

In this project we aim to solve FDE on metric graphs using PINNs. We explore a wide variety of methods to optimize and reduce the errors of the output. We find the L2 errors on manufactured solutions to understand the effectiveness of our neural network.

Parabolic Problem:

Problem 1:

$${}^CD_{0,t}^\alpha u(x,t) + f(x,t) + u(x,t)u_x(x,t) - 0.1u_{xx}(x,t) = 0$$

$$0 < x < 1, \quad 0 < t \leq 1$$

Source Term & Conditions:

- $f(x,t) = -\frac{2e^{xt^2-\alpha}}{\Gamma(3-\alpha)} - e^{2x}t^4 + 0.1e^xt^2$
- $u(x,0) = 0$
- $u(0,t) = t^2, \quad u(1,t) = et^2$

Exact Solution:

For $\alpha \in (0,1)$:

$$u(x,t) = e^xt^2$$

Run 1 Parameters:

Fractional Approximator:L1

Constraint type: soft

Spatial points:100

Temporal points:100

L1 points:0

Iterations:10000

Architecture:60x4

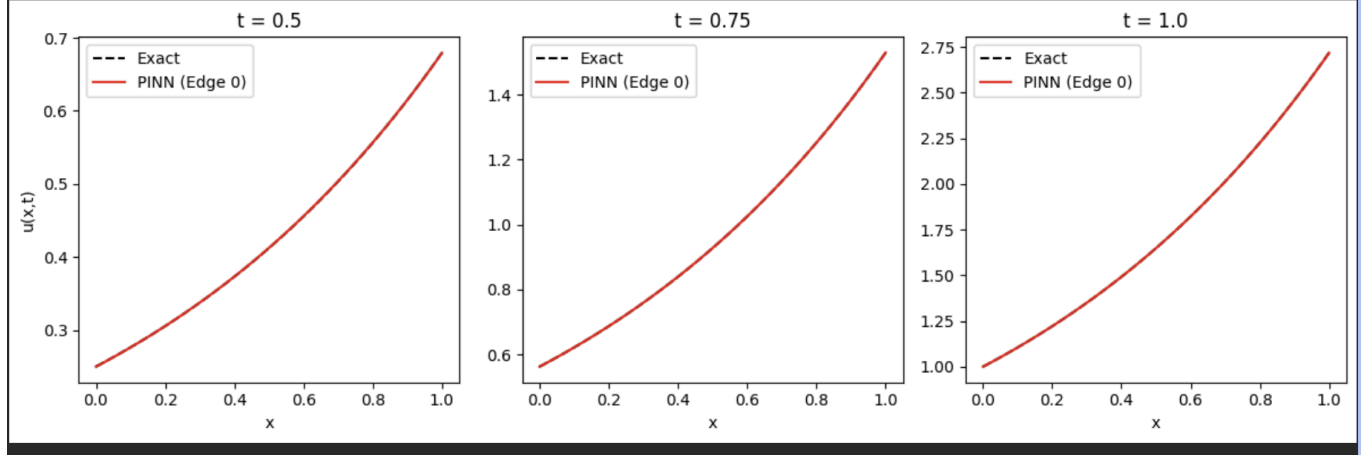
Strategy: Dual

Causal Training: Disabled

NTK balancing: 200 iterations

RAD sampling: Disabled

t=1.0000: 1.4440e-04
Mean: 1.4440e-04



Run 2 Parameters:

Fractional Approximator:L1

Constraint type: hard

Spatial points:100

Temporal points:100

L1 points:0

Iterations:10000

Architecture:60x4

Strategy: Dual

Causal Training: Disabled

NTK balancing: 200 iterations

RAD sampling: Disabled

t=1.0000: 3.5987e-05
Mean: 3.5987e-05

